ASSIGNMENT 2

FINANCIAL ECONOMETRICS

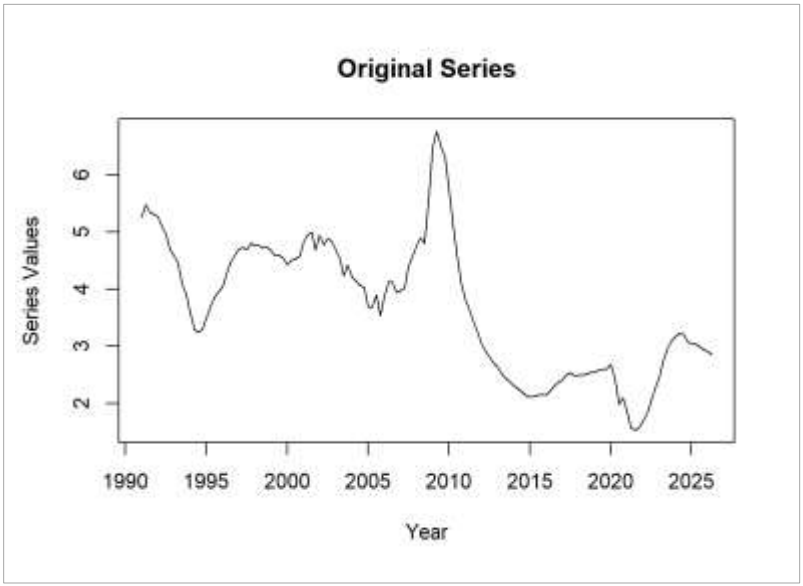
George Johnson

You can work in groups of up to three students. Write all names here. Submit only one report per group. Submit the report in PDF format.

CREDIT CARD DELINQUENCIES

LEVEL SERIES PLOT

Use this text box to describe/discuss the time credit card delinquencies **level** series, what you observed, any thoughts on driving patterns, etc. Use the large box to the right to paste the level series chart.



ADF TEST FOR LEVEL SERIES

Use this box to describe the results of the ADF test for the credit card delinquencies **level** series. Paste the image of the ADF test result in the smaller box to the right.

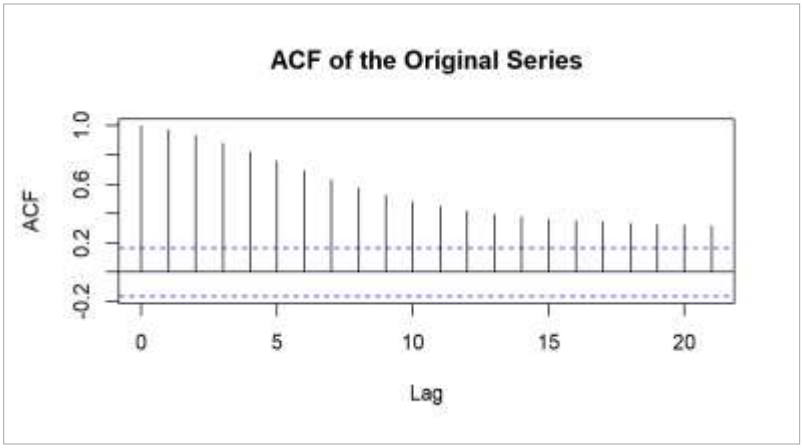
```
ADF Test for the Original Series

        Augmented Dickey-Fuller Test

data:  data$CCDQ
Dickey-Fuller = -1.3791, Lag order = 0, p-value = 0.8349
alternative hypothesis: stationary
```

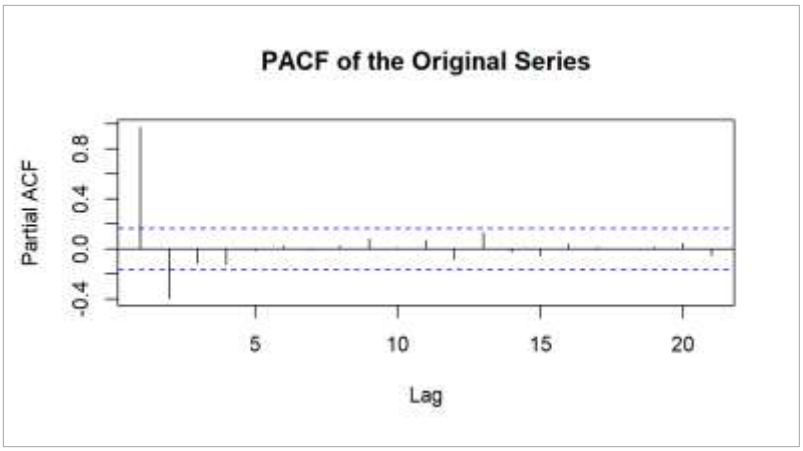
ACF AND PACF

Use this box to describe the ACF and PACF outputs for the credit card delinquencies **level** series. Paste the ACF and PACF outputs in the two bottom boxes to the right.



OTHER OBSERVATIONS

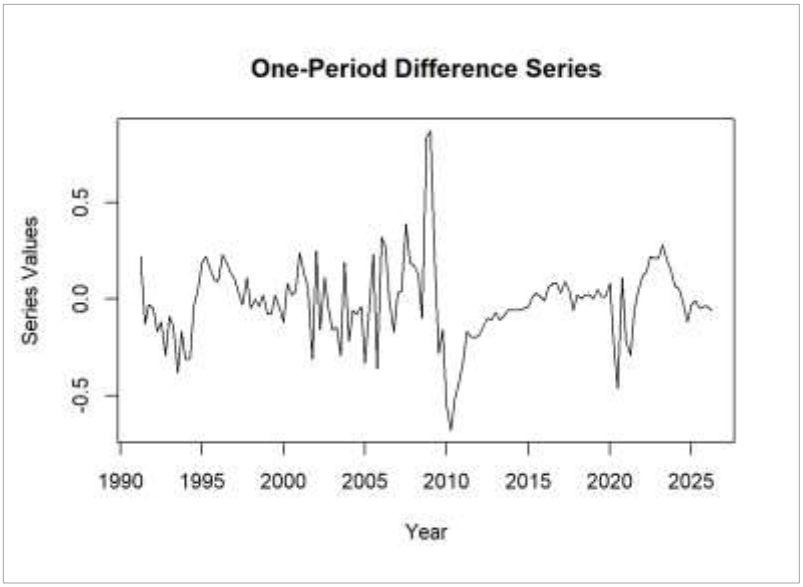
Use this box to describe important observations on the **level** series. Here, explain why you think these analyses are relevant, why do they matter? As you get the series ready for modeling and forecasting, use this box to point out any relevant observations from the outputs on this page.



CREDIT CARD DELINQUENCIES

ONE-PERIOD DIFFERENCE SERIES PLOT

The series fluctuates around a mean of about 0 with no noticeable trend. Volatility is not constant. The 1990s are normal, with the largest changes being seen in 2008 and 2010, a steep rise and decline as delinquencies climbed through the financial crises. A sharp negative happens in 2020, this was most likely due to the stimulus package from covid, with activity rising in 2022.



ADF TEST FOR ONE-PERIOD DIFFERENCED SERIES

The null is that the series is nonstationary. It is rejected, with a Dicky Fuller stat of -6.6586 and a p-val of 0.01, with a warning that the true value is below the floor. Difference once proved stationary, so a second diff is not needed and no seasonal diff needed.

```
ADF Test for the Differenced Series

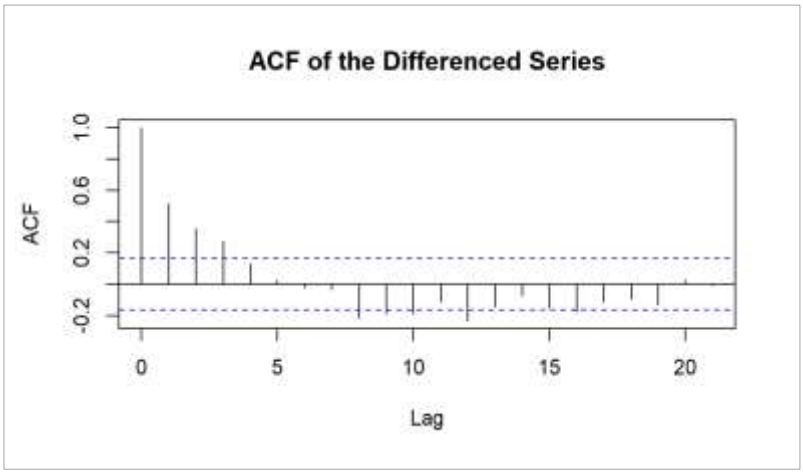
        Augmented Dickey-Fuller Test

data:  dataSCMSCCDQ
Dickey-Fuller = -6.6586, Lag order = 0, p-value = 0.01
alternative hypothesis: stationary

Warning message:
In adf.test(dataSCMSCCDQ, k = 0) : p-value smaller than printed p-value
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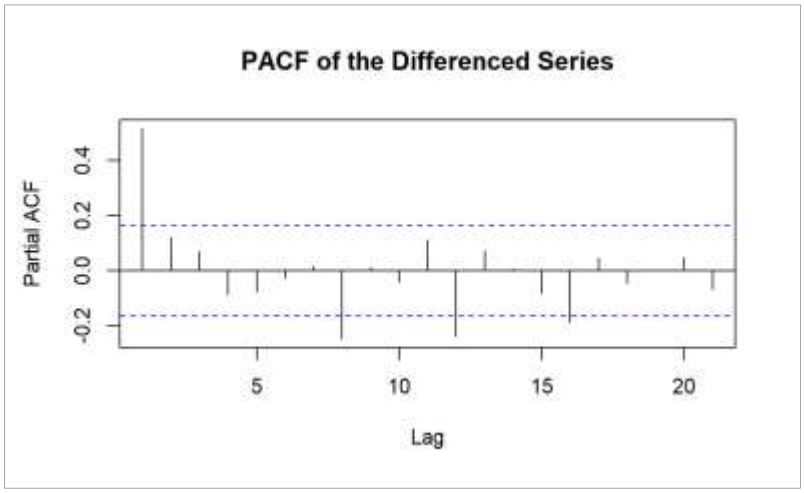
ACF AND PACF FOR ONE_PERIOD DIFFERENCED SERIES

The ACF has significant values at lag 1, 2, and 3 after which it falls within the confidence bands and hovers there majority of the time confirm it becomes stationary. The PACF has 2 significant spikes at lag 1 of 0.5 and then cuts off, with negative exceptions seen at lag 8 and a borderline at lag 12. An ACF that decays with a PACF that cuts off is common in the AR(1) supporting ARIMA(1.1,0) on the level series.



OTHER OBSERVATIONS

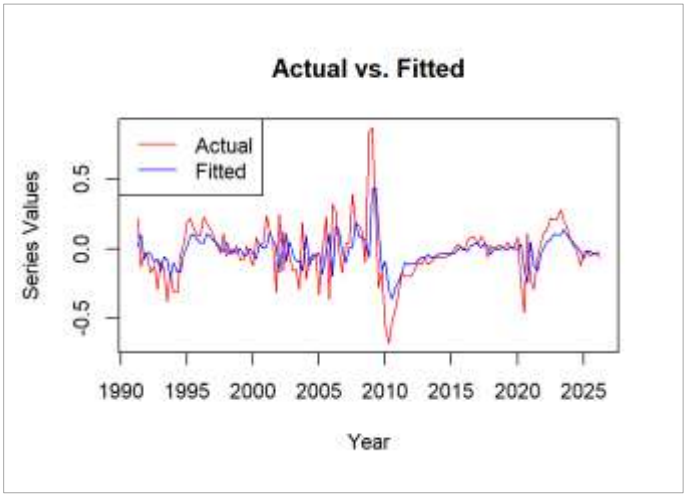
Differncing stabilized the mean but not the variance, with clustering from 2008 →2010 and again in 2020, so the forecast intervals would be to narrow in stressed periods. The negative PACF values at lags 8 and 12 just need to be noted and the model does not to be changed. Lag 4 sits between the bands, and is evidence against quarterly seasonality, and with more than 20 lags tested significance is expected. A Ljung box test would show if anything meangful was unobserved.

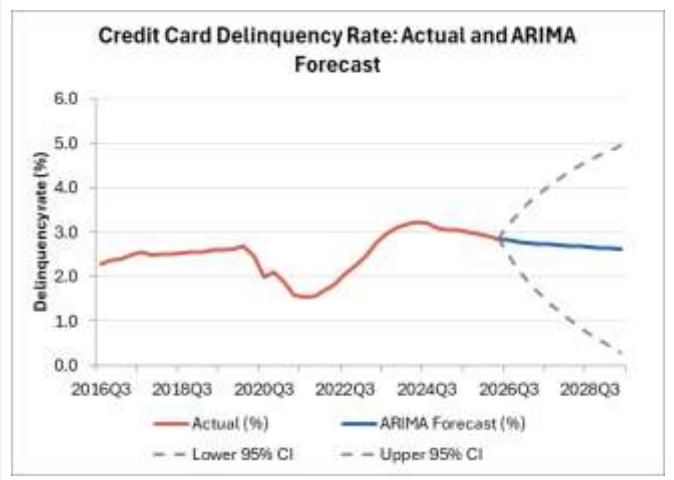


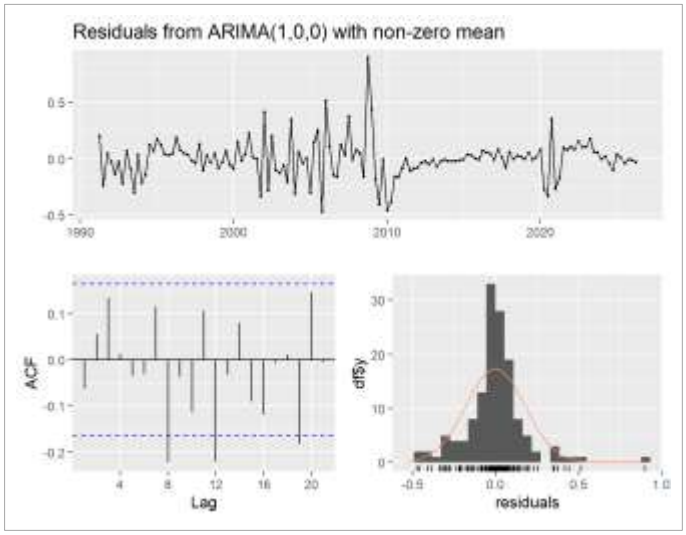
CREDIT CARD DELINQUENCY SERIES, ARIMA FORECAST

CREDIT CARD DELINQUENCY SERIES, ARIMA FORECAST OUTPUT

The model is fitted as ARIMA(1,0,0) on the differenced series, which is the same model as ARIMA(1,1,0). The drift on the level series and the forecast are generated 12 quarters ahead. I fitted on the level with d = 1 which let R undo the differencing, and outputting a percentage. The actual vs fitted shows the model tracking direction while consistently understating the amplitude of large moves. This is expected from an autoregressive parameter. The forecast itself is about flat, easing from about 2.8% toward 2.5% since the last 3 years.







FORECAST OBSERVATIONS

12 quarters is a long horizon in this specification. The AR(1) coeffeciant tells us the forecast reverts towards its drift adjusted past within a few quarters, so almost everything past the first year is the modelrestating the long run tendency not using the information from the data. This is seen in the chart, where the forecast line is nearly flat and 95% of the band spanning 1 to 5 percent by 2028. The parameter choice is support by correlograms, since the PACF cuts off after lag 1 and the ACF decayed, but a defensible horizon here would be four quarters not twelve.

```
        Ljung-Box test

data:  Residuals from ARIMA(1,0,0) with non-zero mean
Q* = 13.527, df = 7, p-value = 0.06026

Model df: 1.   Total lags used: 8
```

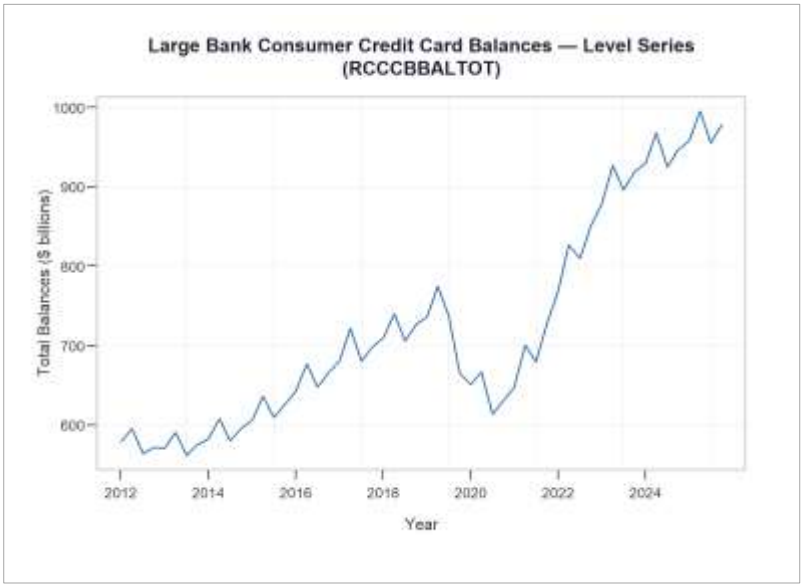
STATEMENT ABOUT MODEL'S FINAL RESIDUALS

The Ljung Box test gives us a Q* of 13.527 on 7 df with a p-val of 0.0606. Wit this we fail to reject the null the the residuals are white noise. The model captures autocorrelation, though the result sits close to the threshold and needs to be noted not rather than it being a clean pass. The residual ACF supports this decision, with bars inside the bands and a few approaching near seasonal lags and large errors spanning from 2008 to 2010.

CREDIT CARD BALANCES

LEVEL SERIES PLOT

Total balances hover around 580 billion in 2012 to 990 billion by 2025. Two things standout, the sawtooth pattern, where balances are built into the fourth quarter then fall back (holiday spending cycle). The second is the 2020 break, where balances dropped 160 billion dollars in a few quarters because of stimulus payments and reduced spending.



ADF TEST FOR LEVEL SERIES

The null is that the series is nonstationary. The Dickey Fuller stats of -1.9613 at lag 0 and a p-val of 0.5902, it cannot be rejected. The level series has a unit root and is ot usable. This is expected because of the upwards trend, and establishes that differencing is required before ARIMA can be fitted.

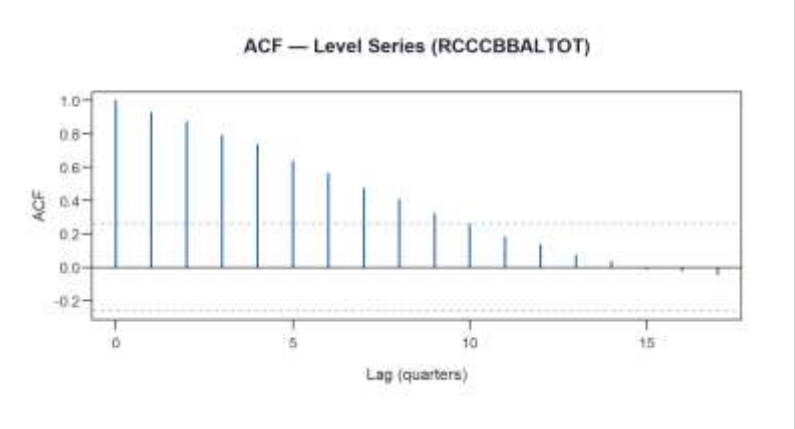
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================================================================
  ADF TEST — LEVEL SERIES (RCCCBBALTOT)
================================================================

        Augmented Dickey-Fuller Test

data:  Original_CCBAL_Series
Dickey-Fuller = -1.9613, Lag order = 0, p-value = 0.5902
alternative hypothesis: stationary
```

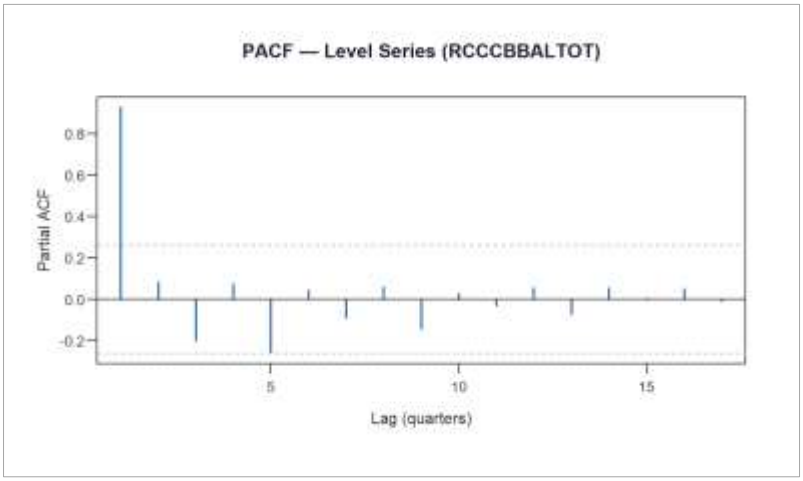
ACF AND PACF

The ACF declines slowly from 1 and stays above the significance band through lag 10. The PACF is dominated by a singular large spike at lag 1 being about 0.9, with the remaining lags having small inside bands a with a few larger negative spikes at lag 3 and 5. A near unity partial autocorrelation and a slow decaying ACF points to a unit root not a stationary autoregressive process.



OTHER OBSERVATIONS

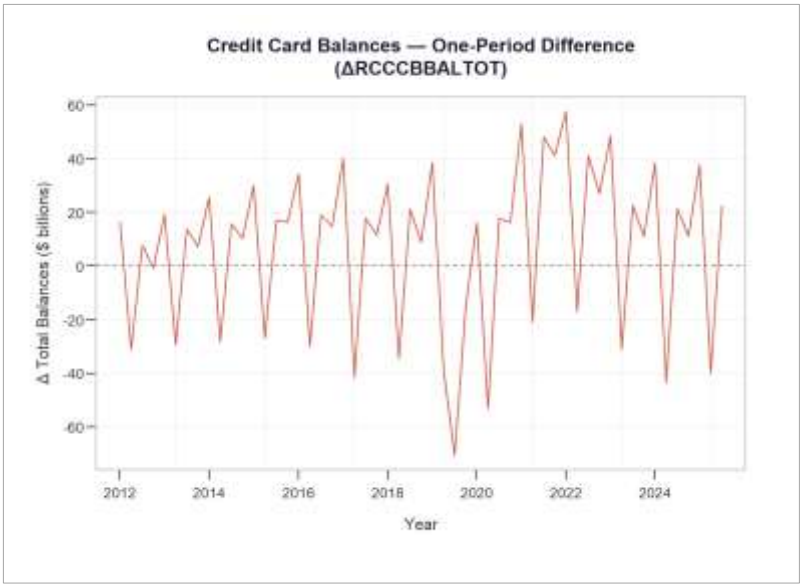
The seasonality visible in the plot does not appear in these correlograms, because the slow decay of the ACF is dominant and hides the quarterly pattern. The pattern should surface once the trend is differenced. Separately, the sample begins in 2012 and covers about 14 years, which is short for a seasonal model. After a regular and sasonal difference are applied the usable observation count falls, so parameter estimates carry more uncertainty then a longer series.



CREDIT CARD BALANCES

ONE-PERIOD DIFFERENCE SERIES PLOT

The first diff removes the trend however it produced a series that oscillates violently between -60 and +60 billion dollars. It is the holiday seasonality, which first diff is unable to remove because it happens at a 4 quarter frequency not 1. The 2020 interruption is the largest most likely due to COVID. The series detrended but is not ready to be modeled.



ADF TEST FOR ONE-PERIOD DIFFERENCED SERIES

With the provided Dickey Fuller test we do not reject the nonstationary null. One diff was not enough because of the seasonal component needing a regular difference. A seasonal diff at lag 4 would be the next step.

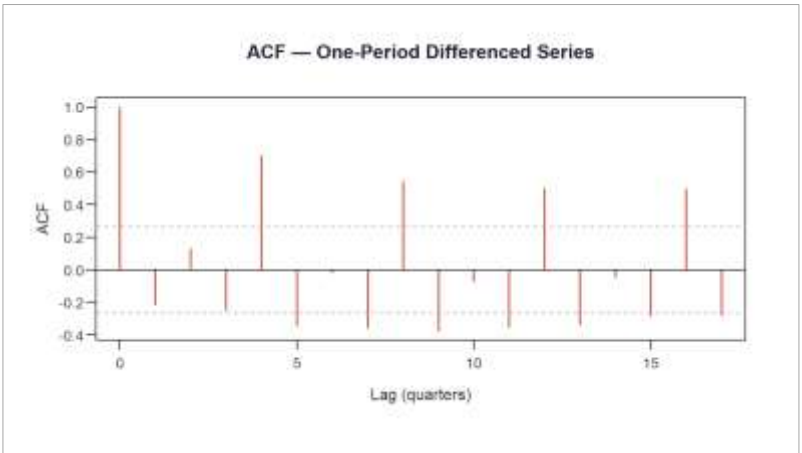
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================================================================
  ADF TEST — ONE-PERIOD DIFFERENCED SERIES
================================================================

        Augmented Dickey-Fuller Test

data:  Differenced_CCBAL_Series
Dickey-Fuller = -1.6397, Lag order = 3, p-value = 0.7198
alternative hypothesis: stationary
```

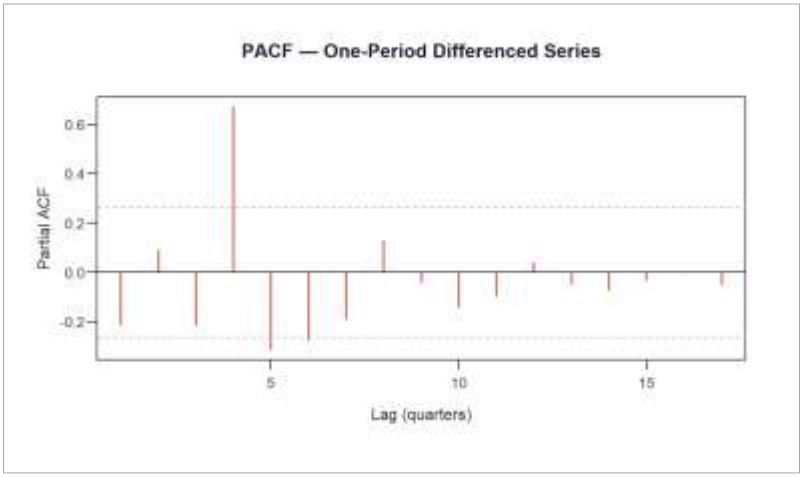
ACF AND PACF FOR ONE_PERIOD DIFFERENCED SERIES

Both of the correlograms show seasonality. The ACF alternates and produces large lags at 4,8, 12, 16 with each running well outside the band. These spikes are evidence of an annual cycle that has not been resolved, and they do not decay showing seasonality. The PACF has one positive spike at lag 4. With both of these test we see a seasonal difference is necessary.



OTHER OBSERVATIONS

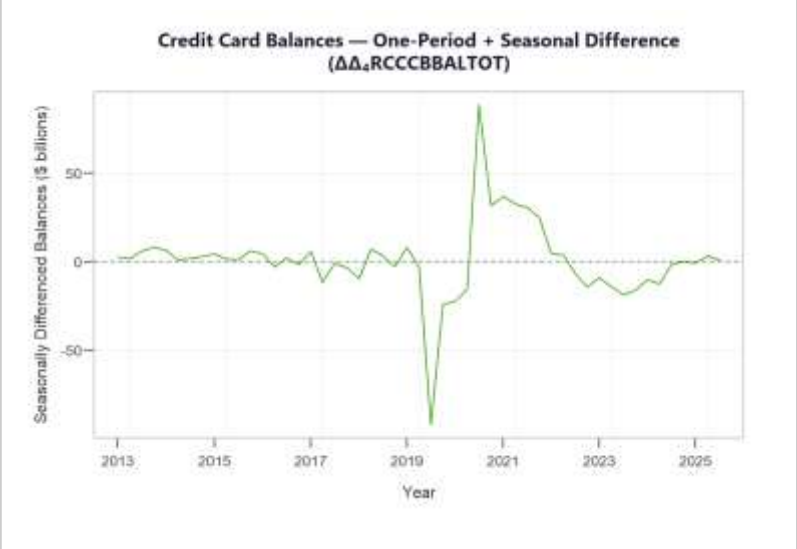
From this section we get the clearest picture that an ADF test alon is not sufficient. The test told us it was nostationary but did not tell why, while the correlograms identified the reason. It also showed why the delinquency series and balance series need different treatment even if they measure similar items, since one is already seasonally adjusted and one is not.



CREDIT CARD BALANCES

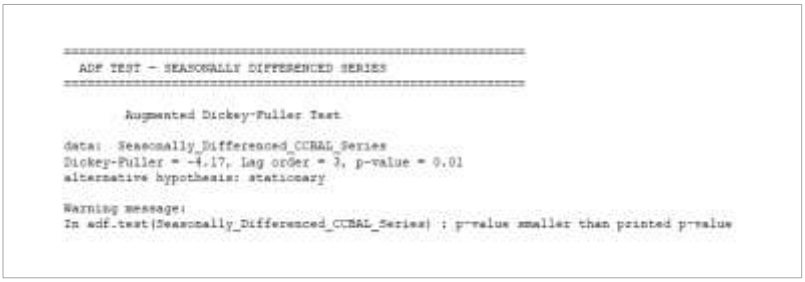
SEASONALLY DIFFERENCED SERIES PLOT

Applying the seasonal diff with the regular diff produced a series that sits close to 0 with no trend and no repeating cycle. The 2020 interruption can be attributed to covid stimulus. Elevated values begin to trend down and return to normal around 2023. Outside of this spike the graph is well behaved.



ADF TEST FOR SEASONALLY DIFFERENCED SERIES

Using the Dickey Fuller test seen in the image we reject the null of nonstationarity. The one regular diff and seasonal diff achieved stationarity.



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================================================================
  ADF TEST - SEASONALLY DIFFERENCED SERIES
================================================================

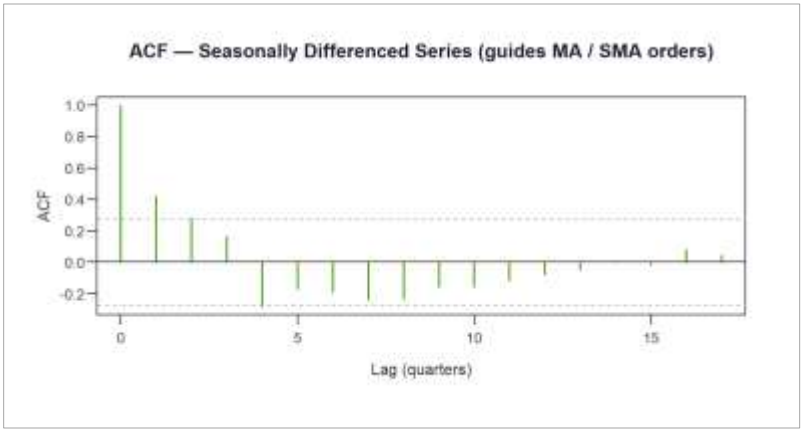
        Augmented Dickey-Fuller Test

data:  Seasonally_Differenced_CCBAL_Series
Dickey-Fuller = -4.17, Lag order = 3, p-value = 0.01
alternative hypothesis: stationary

Warning message:
In adf.test(Seasonally_Differenced_CCBAL_Series) : p-value smaller than printed p-value
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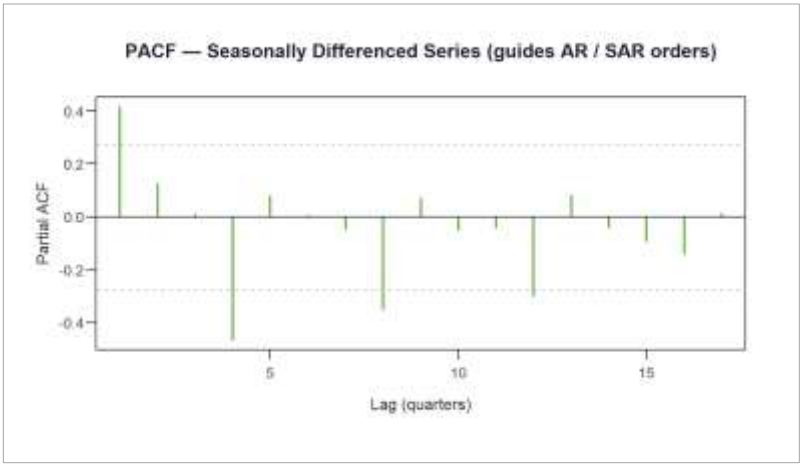
ACF AND PACF FOR SEASONALLY DIFFERENCED SERIES

On the nonseasnal side the ACF decays from 0.38 at lag 1 while the PACF has a positive spike at lag 1, giving evidence for an autoregressive term. Seasonally the ACF shows a negative spike at lag 4 while PACF has negative values at lag 4,8, and 12 that decay across seasonal lags. The negative ACF spike and decaying PACF value represents a sesonal moving term.



OTHER OBSERVATIONS

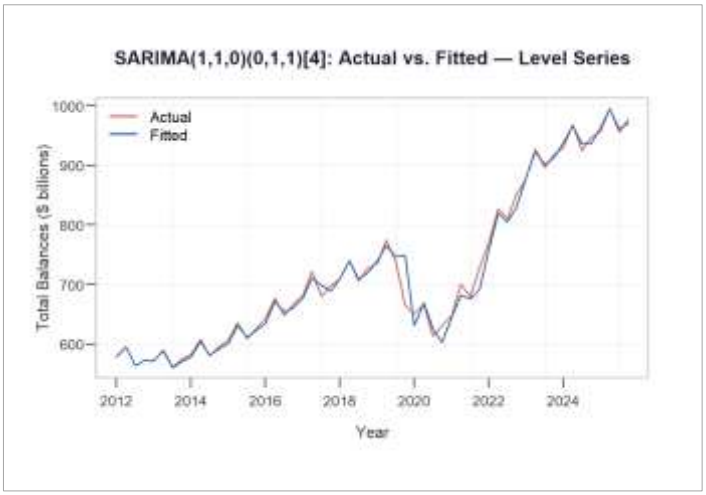
The correlograms support SARIMA(1,1,0)(0,1,1) at period 4. The nonseasnal autoregressive term comes from PACF spike at lag 1 and the seasonal moving average comes from the negative ACF spike at lag 4. The 2020 swing is big enough to influence parameter estimates, and because its in a small sample it does have larger leverage.

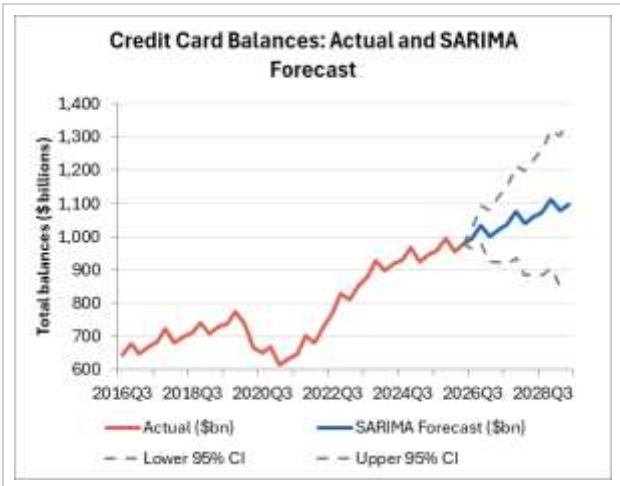


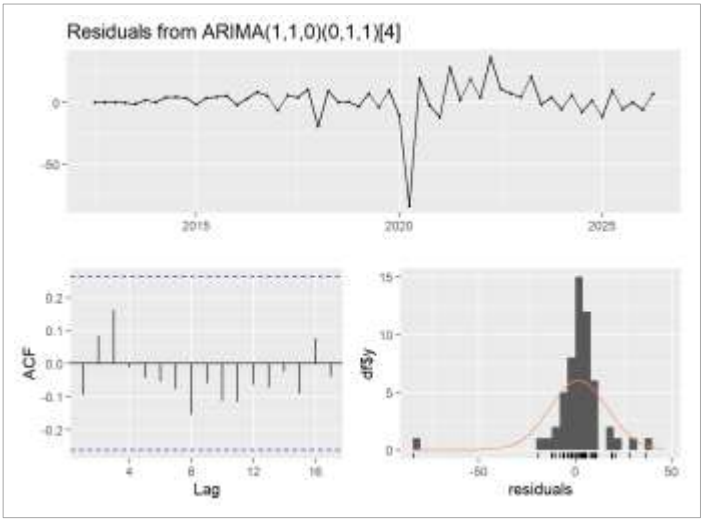
CREDIT CARD BALANCES, SEASONAL ARIMA FORECAST

CREDIT CARD BALANCES SERIES, SEASONAL ARIMA FORECAST OUTPUT

The model was fitted as SARIMA(1,1,0)(0,1,1) at period 4 on the level series, with d and D parameter in utilized in R to apply both differences before the estimation. The forecast is seen in billions of dollars. The actual vs fitted shows close tracking across the full sample, including seasonal peaks and the 2020 shock. The 12 quarter forecast continues up from 990 billion towards approximately 1.1 trillion by 2028, holding seasonal shape, with 95% bands spanning 850 billion to 1.320 trillion at the end of the horizon.







FORECAST OBSERVATIONS

The parameter choice follows the correlograms, with the autoregressive term supported by the PACF spike seen at lag 1 and the seasonal moving term noted in negative ACF spike at lag 4. The forecast also holds the seasonal pattern. 12 quarters is a long projection from a sample that has only 14 years of data, and the band width allows us to see that, since a range of 850 billion to 1.320 trillion is too wide.

```
        Ljung-Box test

data:  Residuals from ARIMA(1,1,0)(0,1,1)[4]
Q* = 4.9356, df = 6, p-value = 0.5521

Model df: 2.   Total lags used: 8
```

STATEMENT ABOUT MODEL'S FINAL RESIDUALS

From the results of the Ljung box test we fail to reject the null that residuals are white noise. The model captures the autocorrelation including the seasonal component, and the residual ACF confirms this with all bars being within the confidence bands. There is only 1 negative outlier in 2020 which can be explained by an economic shock.

DISCUSSION

OUTPUTS AND INTERPRETATION

You are presenting your credit card balance forecast to the bank's retail lending committee. A senior risk officer interrupts you mid-presentation and says, *"The credit card balance projections do not show any sudden change in direction, which suggests consumers are not changing their credit habits. Shouldn't we take that as a warning sign and tighten our underwriting standards now, before delinquencies spike? Are these results somewhat concerning? How confident are you really about this model? What lending strategy should we implement based on the results?* How do you respond? Consider the modeling methodology and results to answer the risk officer's questions and assertions.

The smoothness is a propert of the model not a finding about consumers. A SARIMA extrapolates the autocorrelation estimated from history, it cannot produce a turning point unless something in the structure implies one. We cannot read the absence of sudden change as evidence that credit habbits are stable because that would treat mechanics as data. Another issue is that the model has no behavioral content. There is no unemployment rate, interest rates, underwriting standards, and credit scores. Since it only knows the past balance it cannot warn us about shifting credit habbits because there is no variable carrying this information.

The diagnostics are strong. The Ljung Box p-val is 0.5521, the residuals are white noise, and the fitting values do track the actual series closely. This describes how well the model explain 2012 through 2025, not how well it will forecast. The confidence bands are the real answer. The point forecast is one path and by 2028 the 95% CI spans 950 1.320 trillion USD. That range is what would be asked to held in mind.

Strategy wise, I would not set an underwriting policy from this model. It has a reasonable basline for balance volume so it is useful for funding and capacity planning. Underwriting decisions would need a loss side, which means delinquency work at minimum and realistically model it with macro drivers.

After the presentation, the bank's CCO says, *"We should use your delinquency forecast in our next 10Q report. But I'm worried about two things. It does not seem the forecast is consistent with the period of high interest rates, which historically correlates with higher delinquencies. Don't you think the model is less pessimistic than it should be? Does that mean we should simply override your model's forecast with a more pessimistic number to account for the current interest rate environment? Or should we report the model output as-is and add a qualitative caveat?"* How would you reply to the COO? Justify your answer.

You are right that the model is less pessimistic than rate environments warrant, and right about why. The model has no interest rate variable. The model only knows past behavior of the delinquency rate, so a tightening cycle enters the forecast only to the extent it moved the historical series.

I would not override the number, and the reason matters more in 10Q than it would internally. Once the model output is replaced with a judgement based figure, that number does not have the models support, and the disclosure would have to say it came from judgement. Attributing this to a model would be a disclosure problem not a model preference.

One path we could take would be to report the model output with a qualitative caveat about rate environment, which is honest about what the model does and does not have. Another would be to build a model specification that has the rate variable (an ARIMAX), and report that instead. The last option would be to report the upper band as the adverse case. This is because the model already states its own uncertainty, and the upper band reaches roughly 5% at the far end of the horizon. This would give us a pessimistic figure that is still model derived and defensible.

DISCUSSION

PREDICTABILITY AND APPLICATIONS

A model validator reads your SARIMA model report and sends you a message: *"I ran your model on a loan sample that includes the 2008–2009 financial crisis period. The forecast errors during that window do not seem very large. This is strange and it may suggest the is incorrectly fitted and may need to be redesigned in favor of a macro-factor model that incorporates variables like unemployment, GDP growth, and interest rates. What is your reaction to the validator's assertions?* Justify your answer.

There is a sample problem here. The SARIMA balance is estimated on FR Y-14M data beginning in 2012, so 2008 and 2009 would not be in this sample at all. Running the model against a loan sample in 08 and 09 would test something the regime is not fitted to, and the errors would say little about if the model is correctly fitted. The claim about the errors also is a problem. Small forecast errors through a crisis would be suprsising for any univariate model, since that is where an extrapolative model should fail. A test setup would have to be seen before accepting this premise.

For the macro factor recommendation, I agree with the direction but not the reasoning. A SARIMA has no macro content by construction, so stress testing for a projection conditional on unemployment, GDP growth, or rates, a macro model would be the right tool and not this model. The diagnostics say it is correctly specified, which is extrapolation. Replacing it would change its purpose not correction.

You are explaining the details of your credit card balance SARIMA model to the bank's CFO, who has no modeling background but is known for sharp, practical thinking. The CFO raises two challenges: *"Our loss mitigation team already projects balances and delinquencies to continue increasing. What does your seasonal model add that we're not already getting based on their knowledge of the market?"* The CFO also said, *"Plus, a model validator I spoke with flagged that SARIMA confidence intervals can be too wide. In fact, validators are skeptical about your model."* Respond to the CFO's questions and statements, with an emphasis on whether the model produces reliable forecasts. Justify your answer.

The model adds 3 things that market knowledge does not. A number being the first. Increasing is a direction, while the model does give a magnitude and date, about 1.1 trillion dollars by 2028, which is what funding and capital planning require. The second would be a separation of seasonal movement from underlying growth, and this is the most important. Balances rise in every 4th quarter following the holiday cycle. This matters is whether this years 4th quarter is larger than the seasonal pattern predicts, and the model answers this because it estimates a seasonal component separately. A team working from market knowledge would read normal seasonality as a trend, an error which the seasonal model prevents. The last would be that it is reproducable and auditable. It produces the same answer from the same data no matter who runs it, and states uncertainty. Expert judgement would do neither, which matters if the numbers end up in a filing.

I would push back on the framing. Wide bands are not a defect, theu instead reflect the uncertainty that exist at a three year horizon in a small sample with a structural break. Narrowing them would not make the forecast reliable just elss honest. The validator is pointing to something real however from a different angle. The bands are wide because the horizon is long relative to the sample, and the right answer would be to shorten the hrozion not to tighten the interval.

The model is reliable and within the limits that the diagnostics support. The Ljung Box p-value show residuals are white nosie and that there is a strong fit. Forecast two to four quarters out is defensible.. Three year point forecast would not be, and I would not present them as such. A forecast is reliable within a stated hrozion and stated conditions, not in general.

CONCLUSION

The main lesson would be that the two sereies needed different treatments. The discovery of this is just as important as the result. The delinquency rate arrived seasonally adjusted and needed one diff plus a single autoregressive term. The balance series however required both regular and seasonal differences. The ADF test allowed us to see the balance series failed to reject but did not tell us why, while the correlograms identitfied the seasonality issue. Both models passed their residual diagnostics telling us each described their sample well. It did not tell us either will forecast well, and the large 2020 spike in both is a reminder of what happens when conditions change. Most consequential for how these models get used is that both are univariate. Neither contain unemployment, interest rates, or underwriting standards. They extrapolate structure from 1 sereis which makes them useful for a short horizon baselin for volume planning and unsuitable as the basis for policy or stress testing. The takeaway is to rely on the near term forecast, treat the CI rather than the point estimate as a real ouput, and move to a model with the explanatory variables when the question is coniditonal on the economy.